# Lecture 29

## Section 5.7 (Part II)

### Differential Equations (Continued)

**Example:** Last class, we showed that the fundamental set of solutions of $\mathbf{x}' = A\mathbf{x}$ where

$$A = \begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix}$$

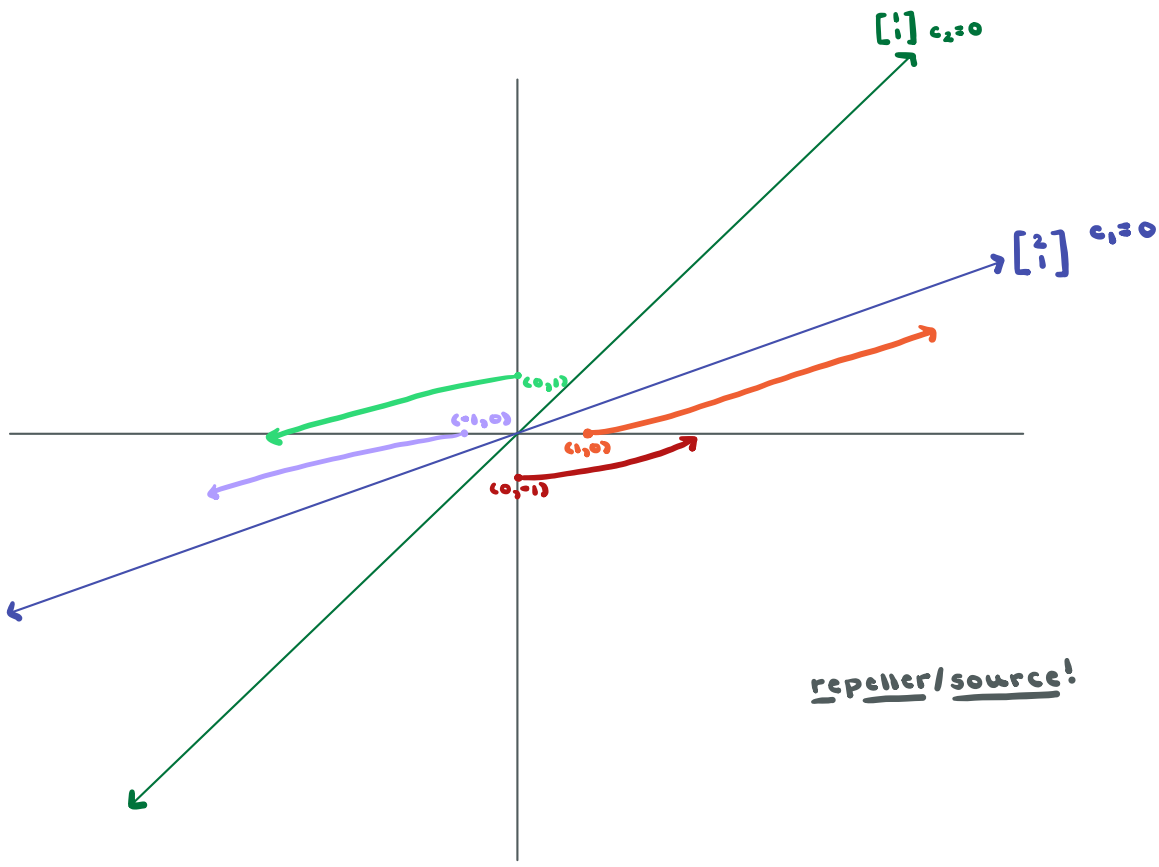
is given by

$$\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t, \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{2t} \right\}.$$

So, a solution $\mathbf{x}(t)$ of $\mathbf{x}' = A\mathbf{x}$ is of the form

$$\mathbf{x}(t) = C_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t + C_2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{2t},$$

where $C_1$ and $C_2$ are real numbers. We can draw the graph, or trajectory, of $\mathbf{x}(t)$ for $t \geq 0$ (varying initial points $\mathbf{x}(0) = \mathbf{x}_0$).



**Remark:** The origin is called an attractor or sink of the dynamical system if all trajectories are drawn into the origin (both eigenvalues are negative). If the trajectories traverse away from the origin, the origin is called a repeller or source (both eigenvalues are positive). The origin is a saddle point when some trajectories approach the origin and then change direction (one eigenvalue is positive and the other negative).

### Decoupling a Dynamical System

Suppose $A$ is an $n \times n$ matrix and $\mathbf{x}' = A\mathbf{x}$. We assume $A$ is diagonalizable, and so $A$ has $n$ linearly independent eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$. The eigenfunctions of $A$ are given by

$$\mathbf{v}_1 e^{\lambda_1 t}, \ldots, \mathbf{v}_n e^{\lambda_n t}, \quad (*)$$

WTS: any solution of $x' = Ax$ is linear combo of $(*)$

where $\lambda_i$ is the eigenvalue corresponding to eigenvector $\mathbf{v}_i$ and $i \in \{1, \ldots, n\}$. Let $A = PDP^{-1}$, where $P = \begin{bmatrix} \mathbf{v}_1 \cdots \mathbf{v}_n \end{bmatrix}$ and $D$ is the diagonal matrix with entries $\lambda_1, \ldots, \lambda_n$. Define $\mathbf{y}(t) = P^{-1}\mathbf{x}(t)$, i.e. $P\mathbf{y}(t) = \mathbf{x}(t)$. Let's substitute $P\mathbf{y} = \mathbf{x}$ into $\mathbf{x}' = A\mathbf{x}$. We see that

$$\frac{d}{dt}(P\mathbf{y}) = A(P\mathbf{y}) = PDP^{-1}(P\mathbf{y}) = PD\mathbf{y}.$$

Note that $\frac{d}{dt}(P\mathbf{y}) = P\mathbf{y}'$, as $P$ is a matrix of constants. This implies that $P\mathbf{y}' = PD\mathbf{y}$, and so $\mathbf{y}' = D\mathbf{y}$. Visually,

$$\begin{pmatrix} y_1'(t) \\ \vdots \\ y_n'(t) \end{pmatrix} = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} \begin{pmatrix} y_1(t) \\ \vdots \\ y_n(t) \end{pmatrix}.$$

So, we have decoupled the system of differential equations, i.e. the derivative of $\mathbf{y}_k$ only depends on $\mathbf{y}_k$. Because $\mathbf{y}_i' = \lambda_i \mathbf{y}_i$, then $\mathbf{y}_i(t) = C_i e^{\lambda_i t}$ for some scalar $C_i$, where $i \in \{1, \ldots, n\}$. Therefore,

$$\mathbf{y}(t) = \begin{pmatrix} C_1 e^{\lambda_1 t} \\ \vdots \\ C_n e^{\lambda_n t} \end{pmatrix}, \qquad \text{where} \qquad \begin{pmatrix} C_1 \\ \vdots \\ C_n \end{pmatrix} = \mathbf{y}(0) = P^{-1}\mathbf{x}(0).$$

If we want to obtain the solution $\mathbf{x}(t)$ for the original system,

$$\begin{aligned} \mathbf{x}(t) &= P\mathbf{y}(t) \\ &= \begin{bmatrix} \mathbf{v}_1 \cdots \mathbf{v}_n \end{bmatrix} \mathbf{y}(t) \\ &= C_1 \mathbf{v}_1 e^{\lambda_1 t} + \cdots + C_n \mathbf{v}_n e^{\lambda_n t}. \end{aligned}$$

### Complex Eigenvalues

Suppose $A$ is a $2 \times 2$ real matrix and $\mathbf{x}' = A\mathbf{x}$. Assume $\lambda$ and $\overline{\lambda}$ are two eigenvalues of $A$, with corresponding eigenvectors $\mathbf{v}$ and $\overline{\mathbf{v}}$, respectively. Two solutions of $\mathbf{x}' = A\mathbf{x}$ are

$$\mathbf{x}_1(t) = \mathbf{v}e^{\lambda t} \qquad \text{and} \qquad \mathbf{x}_2(t) = \overline{\mathbf{v}}e^{\overline{\lambda} t}.$$

**Claim:** $\mathbf{x}_2(t) = \overline{\mathbf{x}_1(t)}$.

*Proof:* Note that $\overline{\mathbf{x}_1(t)} = \overline{\mathbf{v}e^{\lambda t}} = \overline{\mathbf{v}}\,\overline{e^{\lambda t}}$. So, we just need to show that $\overline{e^{\lambda t}} = e^{\overline{\lambda} t}$. We will show that $\overline{e^z} = e^{\overline{z}}$ where $z = a + bi$ for some real numbers $a$ and $b$. Notice that

$$\begin{aligned}
\overline{e^{a+bi}} &= \overline{e^a e^{bi}} \\
&= \overline{e^a}\,\overline{e^{bi}} \\
&= e^a\overline{(\cos(b) + i\sin(b))} && \text{because } e^a \in \mathbb{R} \\
&= e^a(\cos(b) - i\sin(b)) \\
&= e^a(\cos(-b) + i\sin(-b)) && \text{cos is even and sin is odd function} \\
&= e^a e^{-bi} \\
&= e^{a-bi} \\
&= e^{\overline{a+bi}}.
\end{aligned}$$

**Claim:** The real and imaginary parts of $\mathbf{x}_1(t)$ are (real) solutions of $\mathbf{x}' = A\mathbf{x}$.

*Proof:* Notice that

$$\begin{aligned}
\mathrm{Re}(\mathbf{x}_1(t)) &= \frac{1}{2}(\mathbf{x}_1(t) + \overline{\mathbf{x}_1(t)}) = \frac{1}{2}(\mathbf{x}_1(t) + \mathbf{x}_2(t)), \\
\mathrm{Im}(\mathbf{x}_1(t)) &= \frac{1}{2i}(\mathbf{x}_1(t) - \overline{\mathbf{x}_1(t)}) = \frac{1}{2i}(\mathbf{x}_1(t) - \mathbf{x}_2(t)).
\end{aligned}$$

So, since $\mathrm{Re}(\mathbf{x}_1(t))$ and $\mathrm{Im}(\mathbf{x}_1(t))$ are linear combinations of $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$, they must themselves be solutions of $\mathbf{x}' = A\mathbf{x}$.

Let's consider what such solutions actually look like. Suppose $\lambda = a + bi$ for $a, b \in \mathbb{R}$. Then,

$$e^{\lambda t} = e^{(a+bi)t} = e^{at}e^{ibt} = e^{at}(\cos(bt) + i\sin(bt)).$$

So,

$$\begin{aligned}
\mathbf{v}e^{\lambda t} &= [\mathrm{Re}(\mathbf{v}) + i\mathrm{Im}(\mathbf{v})](e^{at})[\cos(bt) + i\sin(bt)] \\
&= [\mathrm{Re}(\mathbf{v})\cos(bt) - \mathrm{Im}(\mathbf{v})\sin(bt)]e^{at} + i[\mathrm{Re}(\mathbf{v})\sin(bt) + \mathrm{Im}(\mathbf{v})\cos(bt)]e^{at},
\end{aligned}$$

implying that

$$\begin{aligned}
\mathrm{Re}(\mathbf{x}_1(t)) &= \mathrm{Re}(\mathbf{v}e^{\lambda t}) = [\mathrm{Re}(\mathbf{v})\cos(bt) - \mathrm{Im}(\mathbf{v})\sin(bt)]e^{at}, \\
\mathrm{Im}(\mathbf{x}_1(t)) &= \mathrm{Im}(\mathbf{v}e^{\lambda t}) = [\mathrm{Re}(\mathbf{v})\sin(bt) + \mathrm{Im}(\mathbf{v})\cos(bt)]e^{at}.
\end{aligned}$$

**Note:** The real and imaginary parts of $\mathbf{x}_1(t)$ are linearly independent functions when $b \neq 0$. So, they will form a fundamental set of solutions of $\mathbf{x}' = A\mathbf{x}$.

**Example:** Suppose

$$A = \begin{pmatrix} 3 & -5 \\ 2 & 1 \end{pmatrix}.$$

We want to find $\mathbf{x}(t)$ such that $\mathbf{x}' = A\mathbf{x}$. The characteristic equation of $A$ is $0 = \lambda^2 - 4\lambda + 13$. One can check that the eigenvalues of $A$ are $\lambda_1 = 2 + 3i$ and $\lambda_2 = 2 - 3i$. The corresponding eigenvectors are

$$\mathbf{v}_1 = \begin{pmatrix} 1+3i \\ 2 \end{pmatrix} \qquad \text{and} \qquad \mathbf{v}_2 = \begin{pmatrix} 1-3i \\ 2 \end{pmatrix}.$$

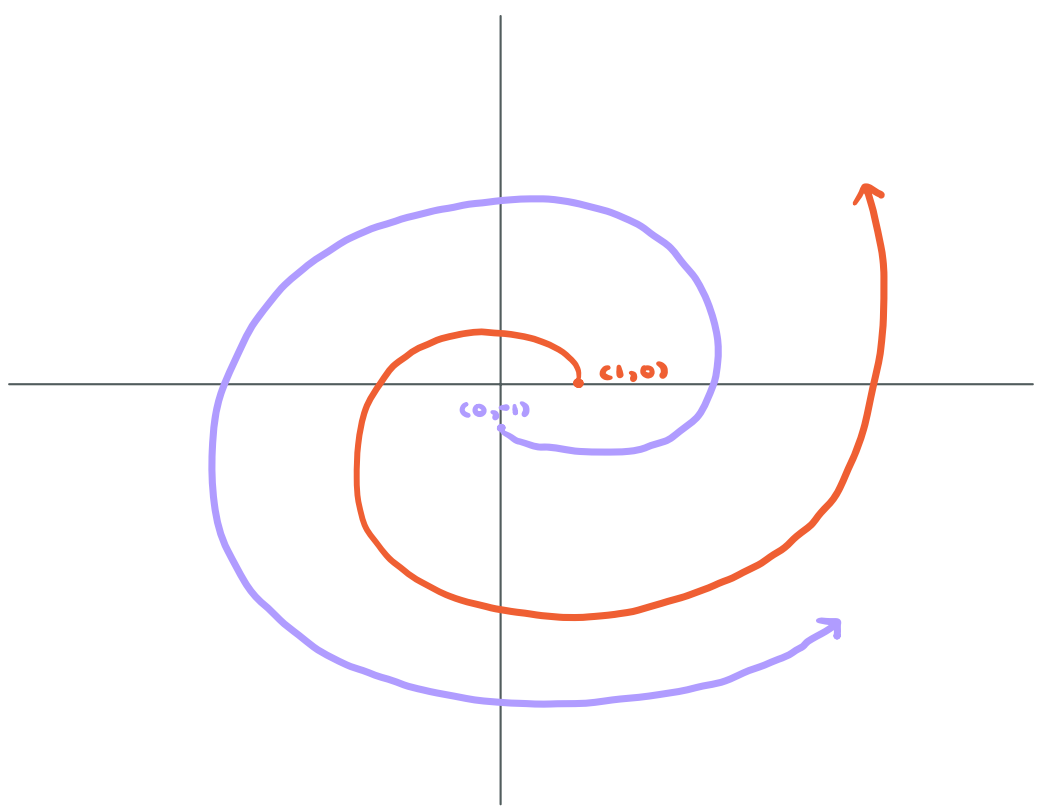
We will find the real and imaginary parts of $\mathbf{x}(t) = \mathbf{v}_1 e^{\lambda_1 t}$ (you can also do this process for $\mathbf{v}_2 e^{\lambda_2 t}$ instead). We see that

$$\begin{aligned}
\mathbf{v}_1 e^{\lambda_1 t} &= \begin{pmatrix} 1+3i \\ 2 \end{pmatrix} e^{(2+3i)t} \\
&= \begin{pmatrix} 1+3i \\ 2 \end{pmatrix} e^{2t} e^{i3t} \\
&= \begin{pmatrix} 1+3i \\ 2 \end{pmatrix} e^{2t}(\cos(3t) + i\sin(3t)) \\
&= \begin{pmatrix} \cos(3t) + i\sin(3t) + 3i\cos(3t) - 3\sin(3t) \\ 2\cos(3t) + 2i\sin(3t) \end{pmatrix} e^{2t} \\
&= \begin{pmatrix} \cos(3t) - 3\sin(3t) \\ 2\cos(3t) \end{pmatrix} e^{2t} + i \begin{pmatrix} \sin(3t) + 3\cos(3t) \\ 2\sin(3t) \end{pmatrix} e^{2t}.
\end{aligned}$$

So,

$$\left\{ \begin{pmatrix} \cos(3t) - 3\sin(3t) \\ 2\cos(3t) \end{pmatrix} e^{2t}, \begin{pmatrix} \sin(3t) + 3\cos(3t) \\ 2\sin(3t) \end{pmatrix} e^{2t} \right\}$$

is a fundamental set of solutions. We graph the trajectories below.



**Note:** In the previous example, the origin is called a spiral point. The rotation is caused by the sine and cosine functions (stemming from the complex eigenvalue). The trajectories spiral outward, since $e^{2t} \to \infty$ as $t \to \infty$.

**Remark:** If $A$ has a complex eigenvalue with positive real part, the trajectories spiral outward. If the real part of the eigenvalue is 0, the trajectories will form ellipses around the origin. If the real part is negative, the trajectories spiral inward.